

June 17, 2026

Dear PhD Admissions Committee,

I am excited to apply for the PhD position "Designing smart space structures for extreme environments: from nonlinear dynamics to resilient mechanisms" at the University of Southampton, located in Southampton, United Kingdom. As a highly motivated and skilled AI Data Science Engineer, I believe my research experience and achievements make me an ideal candidate for this project.

In my previous work on the "Cardiovascular Medical Digital Twin" project, I developed several innovative methods that can be directly applied to the requirements of this PhD offer. Specifically, I would like to highlight two achievements that demonstrate my expertise in nonlinear dynamics and resilient mechanisms: *Medical_{MambaT}S* and *PI_{DeepONet}*.

Medical_{MambaT}S is a linear-time state space model for sub-millisecond bedside inference. This achievement involves processing high-dimensional data and achieving real-time processing, which is crucial for the development of AI-based predictive control methods for resilient space structures. Moreover, *Medical_{MambaT}S*' linear-time complexity ensures efficient computation, a critical aspect in the design of smart space structures.

PI_{DeepONet}, on the other hand, is a 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd.

In addition to these achievements, I possess a strong background in dynamics and control, with skills in numerical modeling and machine learning. I am confident that my unique combination of research experience and technical expertise makes me an ideal candidate for this PhD position.

I am excited about the opportunity to work on this project and contribute to the development of smart space structures for extreme environments. I am confident that my skills and achievements make me a strong candidate, and I look forward to discussing my application with you further.

Sincerely,

Ismail Aissaoui